
Aprendizagem Computacional (AC) Machine Learning (ML)

2026/2027

Practical Exercises :
Week #2

Feature extraction

Feature extraction - Fruits Dataset

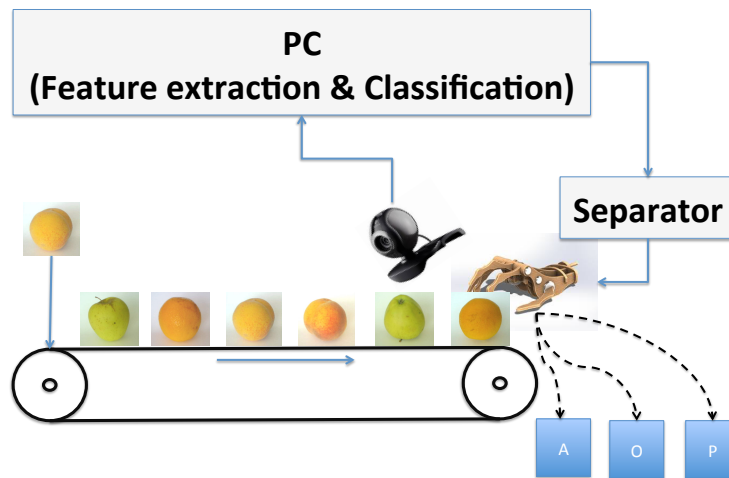


Figura 1: Automatic system to separate fruits in three different containers (classes).

Consider an automatic system, represented in Fig. 1 that aims to separate three types of fruits (apples, peaches or oranges) in different containers. The main system component is a PC equipped with a camera that takes pictures from the fruits in a conveyor belt; extract features from the fruit images; and classifies them in three different classes. In this class we will face the problem of extracting features from fruit pictures. Consider the Fruits dataset available at UCStudent as the pictures taken by the PC camera.

The dataset contains color pictures of 22 apples, 15 oranges and 36 peaches. Each pixel in the pictures is codified in 256 levels RGB.

1. Inspect about possible features.
2. Develop code to extract the features.
3. Plot the features and inspect about how good they are in discriminating the different fruit types.
4. Answer to the following equations:
 - What is the dimensionality of the problem?
 - How many examples are available in the dataset?